

THESIS SUMMARY

Development of a Predictive Clinical Protocol for Early Identification of Multi-Organ Dysfunction Syndrome (MODS) in Post-Operative ICU Patients

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Year: 2024

CGPA: 4.57 / 5.0

BACKGROUND & PROBLEM STATEMENT

Multi-Organ Dysfunction Syndrome (MODS) remains one of the leading causes of mortality in post-operative Intensive Care Unit (ICU) patients. Conventional clinical monitoring systems frequently fail to detect early sub-clinical physiological deterioration, resulting in delayed interventions and significantly worse patient outcomes. The absence of a standardized, predictive clinical protocol for early MODS identification in resource-constrained ICU settings — particularly in sub-Saharan Africa — was identified as a critical gap in critical care nursing practice.

This thesis was motivated by direct clinical observations at the University College Hospital (UCH) ICU, Ibadan, where hemodynamic instability patterns preceding MODS onset were consistently identifiable through careful physiological trend monitoring, yet no formal predictive protocol existed to standardize early detection.

RESEARCH OBJECTIVES

- To identify early physiological markers predictive of MODS onset in post-operative ICU patients
- To develop a time-series analysis framework for monitoring hemodynamic deterioration trends
- To construct and validate a predictive clinical protocol for early MODS identification
- To assess the clinical utility of the protocol in reducing delayed intervention times in ICU settings

METHODOLOGY

A prospective observational study design was employed. Data were collected from post-operative ICU patients at University College Hospital, Ibadan, tracking key hemodynamic and laboratory parameters at defined time intervals.

Key variables monitored included Mean Arterial Pressure (MAP), serum lactic acid levels, heart rate variability, fluid balance, and organ-specific markers. Data were analyzed using:

- Time-series analysis to identify deterioration trends and critical threshold crossings
- Logistic regression modelling (SPSS) to determine predictive variables and their relative weights
- Python (pandas, matplotlib) for data cleaning, exploratory analysis, and physiological trend visualization
- Tableau for dashboard creation and clinical visualization of deterioration trajectories

KEY FINDINGS

- The predictive protocol successfully identified sub-clinical MODS deterioration up to 6 hours ahead of standard electronic alert benchmarks — a statistically significant improvement over existing monitoring systems
- Lactic acid trend elevation combined with MAP decline were the strongest early predictors of impending MODS, with predictive accuracy confirmed by logistic regression analysis

- Heart rate variability and cumulative fluid balance provided secondary confirmatory signals, improving overall protocol sensitivity
- Implementation of early-warning sepsis protocols derived from the framework contributed to a 12% reduction in Code Blue events in the UCH ICU, with interventions initiated 45 minutes earlier on average
- A 15% discrepancy in hemodynamic charting accuracy was identified during data collection, leading to the development of a parallel digital data validation protocol for AI-ready clinical dataset production

CLINICAL IMPLICATIONS

The predictive protocol developed in this thesis offers a replicable, low-resource-intensive framework applicable to ICU settings across sub-Saharan Africa and other resource-constrained environments. The protocol:

- Provides a structured, evidence-based approach to early clinical deterioration detection without reliance on advanced AI infrastructure
- Is directly translatable into nursing practice as a bedside monitoring checklist and escalation trigger
- Establishes a foundation for AI-augmented clinical decision support systems, as the physiological variables and time-series approach are directly compatible with machine learning model inputs
- Contributes to the broader field of clinical AI by demonstrating the value of nurse-led data collection and ground-truth validation in healthcare AI dataset engineering

TOOLS & METHODS SUMMARY

Statistical Analysis: SPSS — logistic regression, time-series analysis

Programming: Python — pandas (data cleaning), matplotlib (visualization)

Visualization: Tableau — physiological trend dashboards

Clinical Domain: Critical Care Nursing, MODS, Sepsis, Hemodynamic Monitoring

Setting: University College Hospital ICU, Ibadan, Nigeria